Test Project Fulltime
New Assistant Recruitment 25-2
Mobile Programming



1. Peserta Test Project Fulltime tidak diperkenankan untuk:
    a. Mempublikasikan jawaban yang telah dibuat secara sengaja ataupun tidak disengaja.
    b. Membuka dan/atau menyalin jawaban dari buku, video, peserta lainnya, atau sumber lainnya.
    c. Melakukan tindakan yang menyebabkan jawaban dicontek oleh orang lain atau kelompok lain, baik disengaja maupun tidak disengaja.
    d. Melakukan tindakan kecurangan lainnya.

2. Jika peserta Test Project Fulltime dan/atau terbukti melakukan tindakan seperti yang dicantumkan pada butir ke-1, maka nilai Test Project Fulltime yang melakukan kecurangan, baik menyontek atau dicontek, akan dinolkan sesuai dengan peraturan yang berlaku. Dan akan diberikan sanksi sesuai prosedur yang berlaku.

3. Jawaban yang dapat diterima dan dinilai adalah jawaban yang dikumpulkan sebelum batas waktu yang telah ditentukan.

TasKalender

Congratulations, Intern! 🎉 You've finally made it into the legendary **TasKalender Inc** company! But hold your excitement, you're not a permanent employee yet. First, you must complete a **quest**: build a mobile application called **TasKalender**.

What is TasKalender? Well, it's not a magical calendar that predicts your future (unfortunately). It's an app where users can log their daily tasks, take notes, and track their income and expenses. Basically, it's a **life organizer**, but with your blood, sweat, and code in it.

Since you are now a **proud** (read: unpaid) member of **Kelson's Kingdom**, your boss has given you a **holy mission**:

1. **Design the ERD (Entity-Relationship Diagram**
2. **Create the Use Case Diagram**

Your **wise** (and totally non-technical) boss has some conditions:

- The database is **your choice** (Firebase? SQL? A notebook under your bed? Up to you!).
- The app **MUST** be built using **Android Studio** with **Kotlin or Java**.
- It must support **Android SDK 33 (Tiramisu)**—because we only serve the freshest versions.
- **MVVM architecture is required**—because spaghetti code is only acceptable in Italian restaurants.
- Be prepared to **explain your code** to your boss, who thinks "clean code" is something you do with a mop.

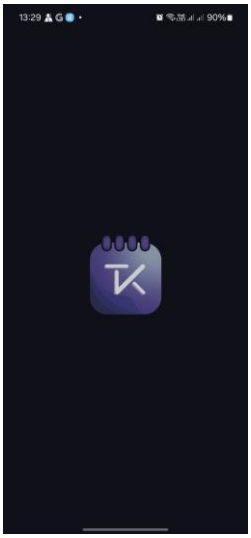
Oh, and about the UI? Your boss already designed one as *inspiration*. He doesn't expect you to copy it pixel by pixel, but **he will judge you** if your design looks like a toddler's art project. Keep it **tidy, responsive, and aesthetically pleasing**—like a well-dressed penguin.

For reference, your boss suggests downloading an app from the Play Store called **"TasKalender"**—created by a previous employee who *mysteriously* left the company. No pressure.

Now go forth, **Intern Extraordinaire**, and prove you have what it takes to earn your spot in Kelson's Kingdom! 🚀

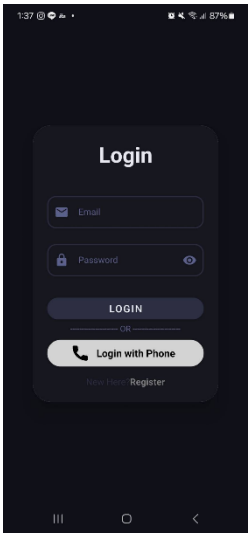
1. **Splash Screen**

   When you first open the application, you must display a **splash screen** with the **logo** on the centre of the screen. After **2** seconds, **automatically redirect** to the **Login Page**.



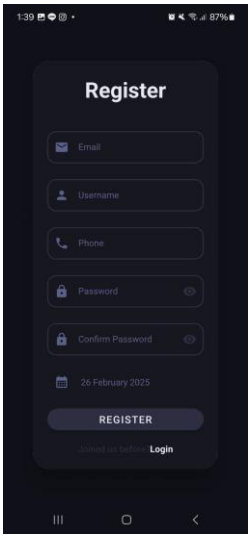
2. **Login Page**

   On this page, the user needs to be able to **log in** using their registered **email** and **password**. The "**Login with Phone Number"** feature **DOES NOT** need to be implemented. Make sure to add a **button** that **redirect**s to the **Register Page**. When user try to log in, **validate** that the **input fields** must **not** be **empty** and **display the error message** if **validation failed**. Don't forget to add a **toggle** to **change** the **visibility** of the **password**. If user has **already logged in** before, then **automatically redirect** them to the **Home Page**.



3. **Register Page**

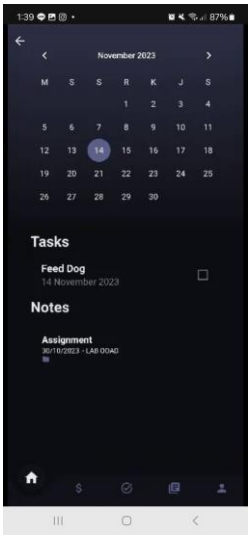
In this page, the user needs to provide information about **email, username, phone number, password, password confirmation, and date of birth.** Make sure **all fields must** be **filled**, **email** must be using **email format**, and **email** has **not** been **used** to make **another account** before. **Display** an **error message** if the **validation** for each field is **not passed**. Don't forget to add a **toggle** to **show/hide** the **password** and the **confirmation password** and a **button** to redirect to the **Login Page**.



4. **Home Page**

   Once you entered the page, there would be a **navigation bar** on the **bottom of the page**. The pages are **Home, Cash, Task, Notes, and Profile**. Make sure to **display** the **selected page** in the **navigation bar**.

   In home page, display a **calendar**, **task**, and **max 5 recently opened notes**. The calendar will act as a **filter** to only **display** the **unfinished task at the selected date**. You can set the **finish** the **task** on this page. By **default**, set the **selected calendar** as **today's** date. The **information** about the **task displayed** is the **name** and the **due date**. The notes would only **act** as a **display** and the user will **not** be **directed** to **notes page** when they **click** on a note at the **Home Page**. Make sure that the display of the notes and task will not overflow vertically, you can try using vertical scroll bar or others android functionality to achieve this.

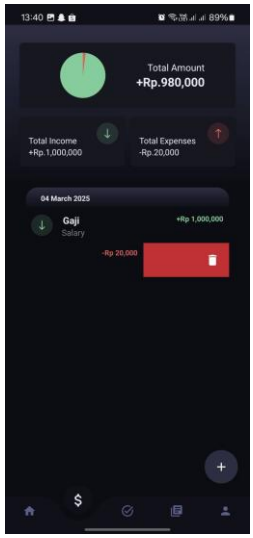


5. **Cash Page**

   In this page, you must **display** the **total** income and expense flow using **a graph/pie chart/chart** in a **month**. Below the chart, you would need to **display** each income and expense **grouped** by the **day** that the **transaction happens**. You could also **delete** the **transaction** by swiping it to the **right** or **left** side.

   At the **bottom right** of the page, there would be a **button** where the user could **add** new **transaction**, when clicked, show a **tab layout** for adding a **new transaction**.

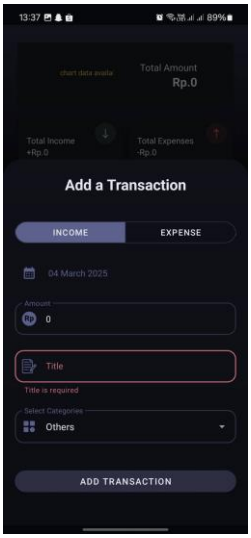


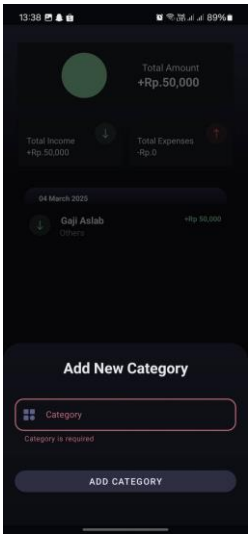


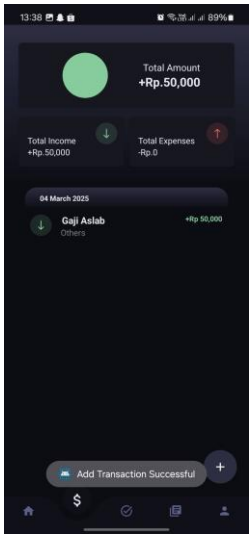
6. **Cash Page (Add New Transaction)**

   Adding new transaction will show as a **tab layout** in the **cash page**. The user could choose either **income** or **expense**. **Validate** the input fields (**amount, date, title, and category**) must **not be empty**, **amount** must **not** be **negative number**, **default date is today**, and **default category is other**. Display **error message** if the **validation is not fulfilled**. The user could also **add** a **new category** in this page. **Validate** the **name** of the **category** is **not empty** and **does not exist yet**. Don't forget to add **success pop up message** if the user

successfully added a new **transaction** or a new **category**. The user could also **close** this **tab layout** by **clicking** the **page outside the layout tab or by swiping it down**.

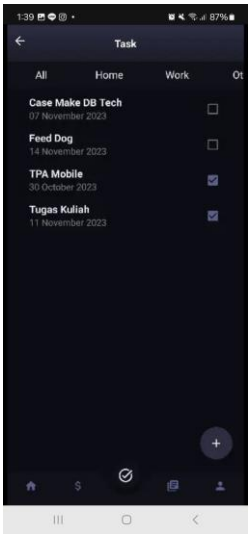


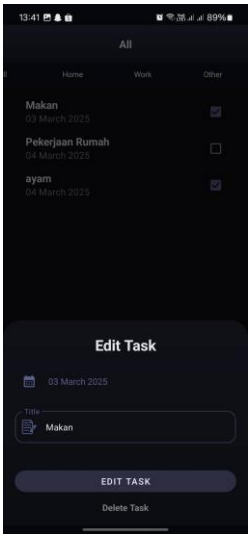




7. **Task Page**

The task categories by default are **All, Home, Work, Other.** The user could **not add** a new **category**. When the user **chooses** a **tab**, only **task** with the **selected categories** will be **displayed**. The user could **finish**, **edit**, and **delete** an **existing task**. When the user clicks the **add button** at the **bottom right** of the page, **show the Add New Task Layout**.

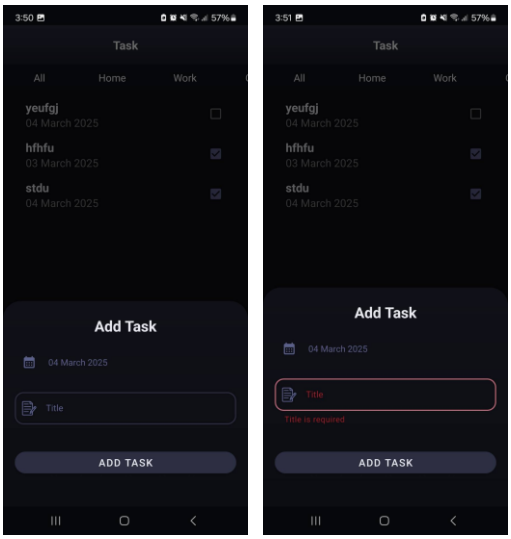




8. **Task Page (Add New Task)**

The user can **close** this **layout** by **clicking outside** of it or by **swiping down**. Ensure that the **task title** is **not empty** and set the **default date** to **Today**. If the user **creates** a **task** while **selecting** a **specific category**, the task will be **assigned** to that **category**. However,

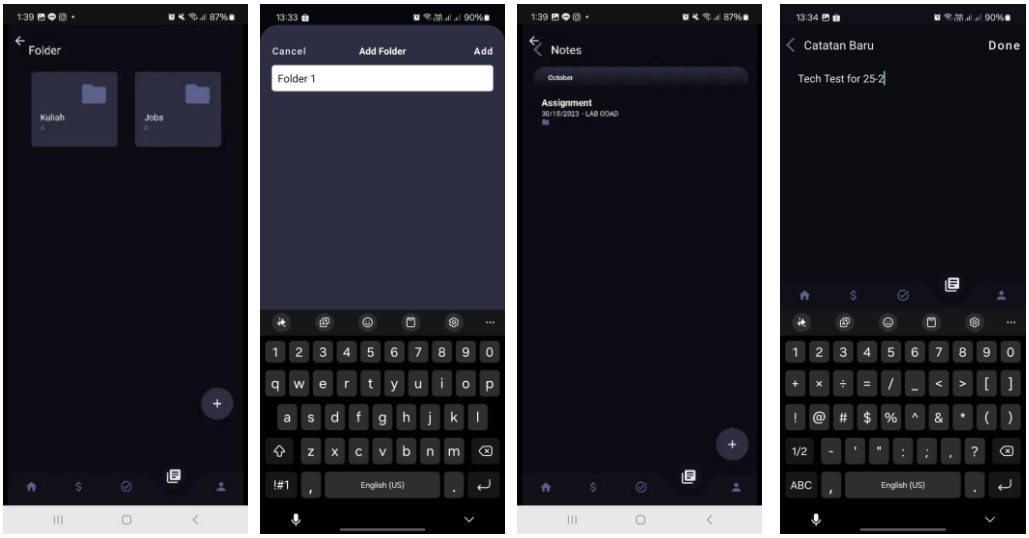
if they **create** a task while the **'All'** category is selected, the **default category** will be **'Others'**.

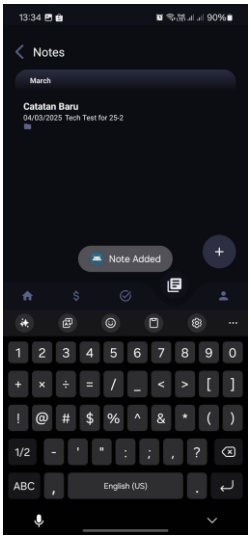


9. **Notes Page**

The **folder name** and **number of notes** inside it will be shown using **list view**. The user could **make new folder** by clicking the **add button**. **Validate** the **name** of the folder must **not be empty** and **unique**.

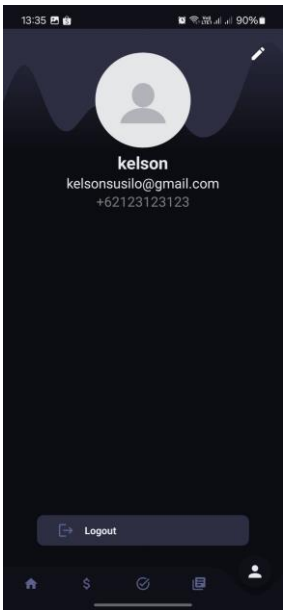
To **make** a **note**, the user must **click** one of the **folders** that they have **made** and **click** the **add** button **inside** the **folder**. **Display** all the **notes group** by the **months** and **previous 30 days from** the **day it's created** in **list view**. The **notes display** must display the **title**, **date created**, and **some preview of the notes**. The **notes** will be **saved** to the **system** when user click "**Done**" **button** at **the top of the page**. Don't forget to **add success message** when the **folder** and the **notes** are **successfully created**.





10. Profile Page

The **username, email**, and **phone number** of the **user** will be **shown** in this page. Add a **Logout button** that if click will **redirect** the user to the **login** page and **prompt** the user to **login again** if they **re-open** the **application**. **Add** an **Edit Profile Button** at the top right of the page.



11. Edit Profile Page

Add a **back button** that will **redirect** the user to the **Profile Page**. In this page, the user could **change** their **email, username, phone number,** and **date of birth**. **Validate** the **newly inserted value** must **not** be **empty**, **must** be **email format**, and **email** must **not** have **used before**. **Display error message** if the **validation** is **not successful**. After **updating** the **profile**, **redirect** user back to the **Profile page** and **display** a **success message**.

The user could also **update** their **password** in this page. **Validate** that the **new password** must **not** be the **same** as the **old password** and the **newly** created **password** must **match** the **password confirmation**. **Display error messages** if the **validation** is **not successful**. After **updating** the **password**, **redirect** user back to the **Profile** page and **display** a **success message**.

